



INTERNSHIP PRESENTATION

The EBVCube Project

Working with netCDF for Essential Biodiversity Variables

A practical introduction — the theory, building an EBV cube from scratch,
and what the `ebvcube` R package can do for your data.

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Why should this matter to you?

Biodiversity data and results are produced everywhere in this institute — but they are hard to find, compare and reuse across studies, scales and different formats.



Hard to find

Results live in scattered files, formats and folders.



Hard to compare

Different grids, time spans and units don't line up.



Hard to reuse

Without shared metadata, others can't build on your work.



Your results can become an EBV cube — findable, comparable and reusable.

Where we're headed



1

The theory

EBV's concept, NetCDF, the scenario/metric hierarchy, and EBV cubes.



2

Create an EBV cube

From metadata JSON to a finished EBV netCDF — in four steps.



3

Work with it

Read, subset, analyse and visualise — a few lines each.



PART 1

The theory

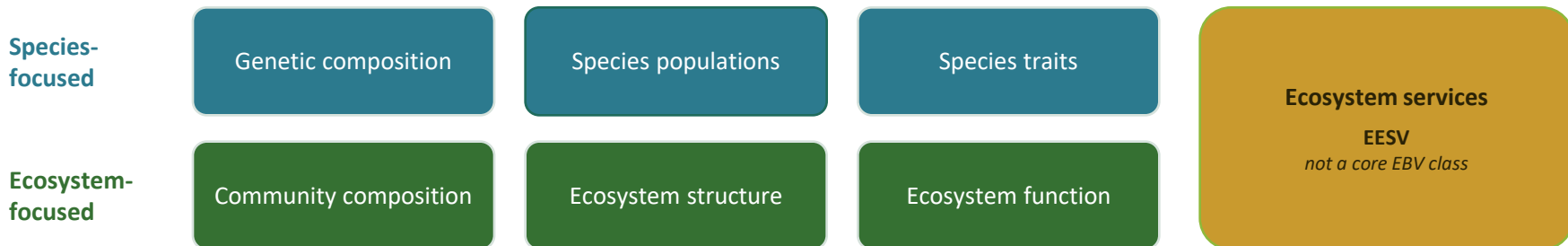
What Essential Biodiversity Variables are, why the community stores them as netCDF, and how an EBV cube is structured inside.

1

What is an Essential Biodiversity Variable?

Defined by **GEO BON**, an EBV is a standardized biological measurement that sits *between* raw observations and the high-level indicators used by policy — with the potential to serve as a foundation for biodiversity monitoring programs around the world.

GEO BON groups the six EBV classes by biological level — species-focused and ecosystem-focused:



Species populations is the class of the example dataset I built (Part 2).

Pereira et al. (2013) - EBV; Balvanera et al. (2022) - EESV



The EBV Data Portal

Every published EBV dataset lives on the **EBV Data Portal**. Each one comes with a stable ID, a JSON metadata record, a downloadable netCDF, and a citation with a DOI.

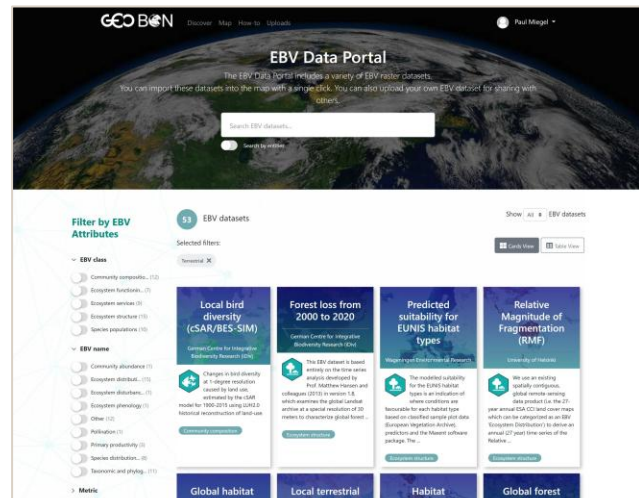
50+

datasets, openly
available today



Pull data straight into R by its
portal ID — no manual download.

→ portal.geobon.org



```
ebv_download(id = NULL,  
outputdir, overwrite = FALSE)
```

List the whole catalogue, or download one
dataset by its ID.

Langer et al. (2022) - EBV Data Portal; Quoß et al. (2024) – EBVCube R package

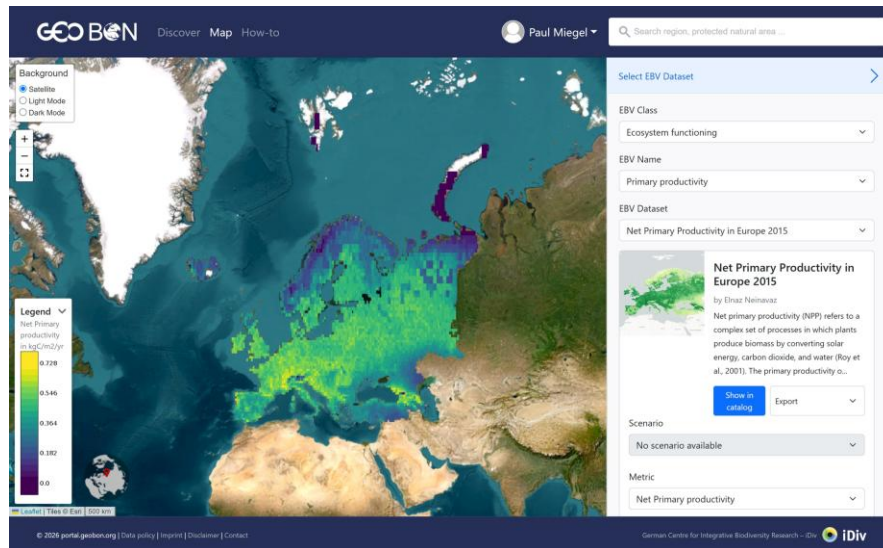
The map viewer



The portal's map viewer is the **fastest way** to explore a dataset — right in the browser, no download required.

- Pick EBV class → name → dataset
- Set scenario, metric & timestep
- Read values off the colour legend
- Export, or open in the catalogue

→ portal.geobon.org/map



Net Primary Productivity in Europe 2015 (ID #59), shown in the portal's map viewer.

Langer et al. (2022) - EBV Data Portal; Quoß et al. (2024) – EBVCube R package

Why netCDF?

Biodiversity data is multi-dimensional — it varies in space, over time and across many entities. netCDF's array-oriented hierarchical structure fits this kind of scientific data perfectly, and one file carries the data and its full description together.



Self-describing

Data and metadata travel together in one file.



Multi-dimensional

Built for array data: space × time × entities.



Standardised

Follows the CF 1.8 and ACDD 1.3 conventions.

One self-describing netCDF replaces a whole folder of loose GeoTIFFs — and never loses the metadata that ties them together.

CF Conventions v1.8 & ACDD v1.3; Quoß et al. (2022)

Hierarchy of the file: scenarios, metrics, cubes

```
root/
├── scenario_1/
│   ├── metric_1/ → ebv_cube
│   └── metric_2/ → ebv_cube
└── scenario_2/
    ├── metric_1/ → ebv_cube
    └── metric_2/ → ebv_cube
...
```

→ <https://github.com/EBVcube/EBVCube-format>

Scenarios (optional, e.g. SSP/RCP pathways) wrap **metrics** (always present), and each metric holds one **ebv_cube**.

The slash-path to a cube is its **datacube path**:

`scenario_1/metric_2/ebv_cube`

`ebv_datacubepaths`(filepath)

List every cube path in a file.

Every EBV cube has four dimensions



lon

Longitude

the x-coordinate of each grid cell



lat

Latitude

the y-coordinate of each grid cell



time

Time

annual, decadal or scenario years



entity

Entity

the species, group or ecosystem
type

Every cube is **[lon, lat, time, entity]**. The entity axis is what makes EBV cubes special — one file can hold hundreds of species on the same grid.

Quoß et al. (2022)

Rich metadata — and how to look at it

ebvcube parses a file's metadata into tidy slots you can read directly:

@general	title, class, creator, license...
@spatial	CRS, extent, resolution
@temporal	start, end, dates
@metric	name, units
@scenario	name (if any)
@ebv_cube	units, fill value, type

```
ebv_properties(filepath, datacube_path = NULL)
```

Read all the metadata of a file or one cube.

Or peek without code:



Panoply

Drag a file in to browse groups, dimensions and attributes, and draw quick maps.



EBVCubeVisualizer for QGIS

Explore slices by time, entity, scenario and metric — right inside a GIS.

*Tools: Panoply (NASA GISS); EBVCubeVisualizer (iDiv)
Quoß et al. (2024) – EBVCube R package*



PART 2

Create your EBV cube

From your own results to a portal-ready EBV netCDF — described, structured and filled, in four steps.

2

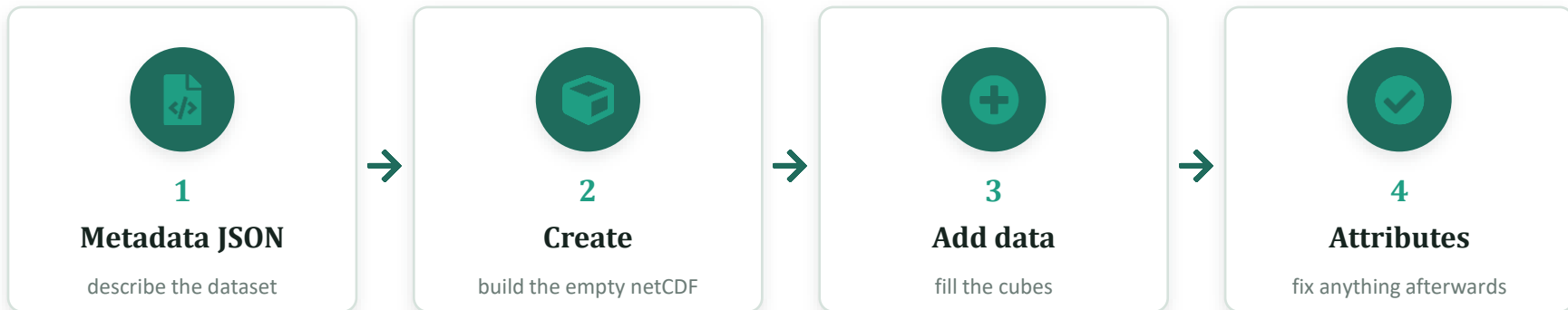
The ebvcube R package

ebvcube wraps the netCDF complexity behind a few clear verbs — **discover**, **read**, **subset**, **visualise** and **create**. It's on CRAN, GitHub and the b-cubed r-universe.

```
install.packages("ebvcube")
```

→ <https://cran.r-project.org/web/packages/ebvcube>

Creating a cube is four steps:



Quoß et al. (2024) – EBVCube R package

Step 1 — describe it: the metadata JSON

Every EBV cube starts from a metadata record. It captures the controlled-vocabulary terms that make the file discoverable:

- EBV class & name
- Entity type & scope (the species)
- Spatial scope, domain, time coverage
- Metric definitions (name, units)
- Creator, license, references

Publishing on the portal is the only way to get a dataset ID.

```
ebv_metadata(outputpath,  
title,  
  ebv_class, ebv_name, ...)
```

Collect every metadata term into one JSON file.



Recommended — use the portal.

The form enforces the vocabularies and is faster than coding the JSON by hand.

Steps 2 & 3 — build it, then fill it

```
ebv_create(jsonpath, outputpath, entities, epsg = 4326, extent, resolution,  
fillvalue)
```

Create the empty netCDF structure that matches your JSON — all dimensions and groups, filled only with the no-data value.

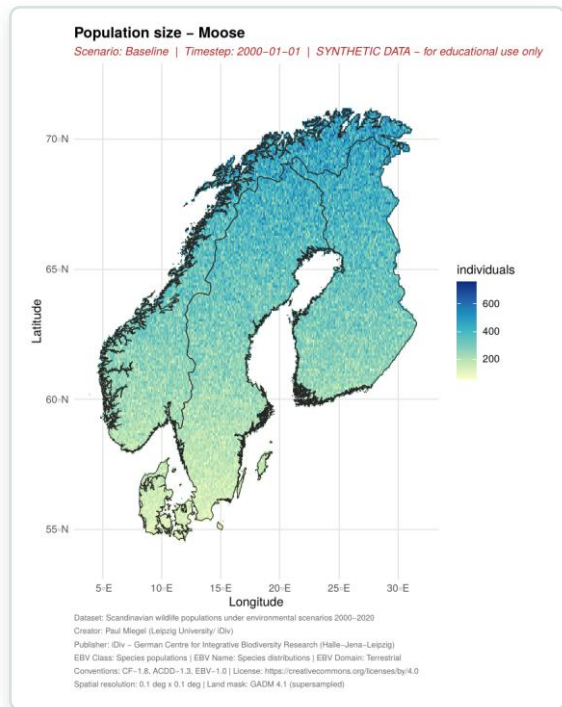
```
ebv_add_data(filepath_nc, datacubeopath = NULL, entity = NULL, timestep = 1, data,  
band = 1)
```

Pour your raster data (GeoTIFFs or in-memory layers) into the cubes — one entity and time slice at a time.



Step 4 — `ebv_attribute()` fixes any wrong unit, title or description in place, without rebuilding the file.

The result: a cube I built from scratch



Moose population size for baseline scenario, first timestep.

Scandinavian wildlife

A synthetic, fully EBV-compliant cube — built end-to-end with the four steps. (educational use only)

- 3 species** Moose · Reindeer · Wolf
- 4 metrics** population size · relative abundance index
habitat range extent · average body weight
- 3 scenarios** baseline · climate change · hunting pressure
- 5 time steps** 2000–2020
- 0.1° grid** GADM 4.1 land mask

Data: GADM v4.1 administrative boundaries



PART 3

Work with it

Reading, subsetting, analysing and visualising EBV cubes — usually just a line or two of code.

3

Reading, subsetting & analysing

Point any reader at a file, pick an entity and time step, and choose how much comes back — the whole cube, a region, or a lazy on-disk view for huge files.



`ebv_read()`

Read a slice into R — array, raster or delayedarray.



`ebv_read_bb()`

Read a subset by bounding box.



`ebv_read_shp()`

Read a subset clipped to a polygon / shapefile.



`ebv_analyse()`

Summary statistics for a cube or region.

Too big for memory? Read it lazily (as `delayedarray`) and only the chunks you touch are loaded.

Quoß et al. (2024) – EBVCube R package

From a quick look to a publication figure



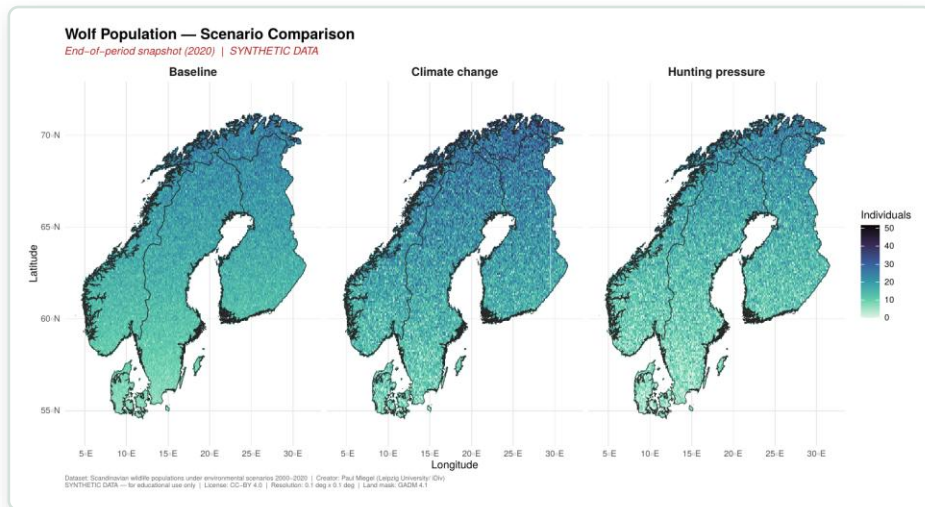
One line for a first look

`ebv_map()` and `ebv_trend()` — an instant classed map or a mean-over-time plot (that also returns the values).



Full control

Read as a `SpatRaster` and pipe it into `ggplot2` + `tidyterra` for publication-quality maps.

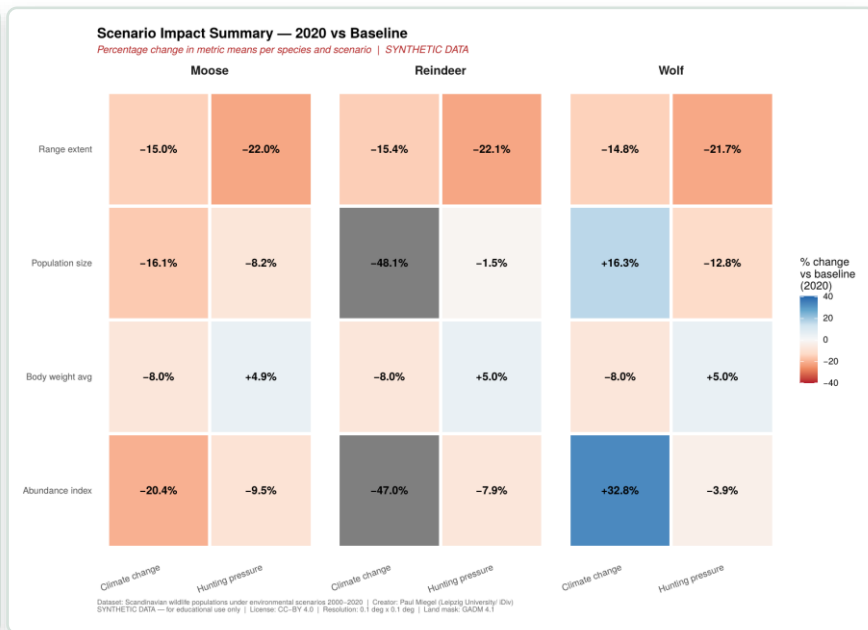


Wolf population across three scenarios — faceted with tidyterra.

Compare species, scenarios and metrics at a glance



Population trends per species & scenario, 2000–2020.



Change in each metric vs. baseline, per species & scenario.

Make your results an EBV cube

If you work with biodiversity data, it can live as an EBV cube — standardised, self-describing, and citable with its own DOI.



Access cubes

Browse, download and cite published EBV cubes — each gets its own DOI.



Learn the format

The EBV netCDF format defined: dimensions, groups and CF/ACDD metadata.



Build with ebvcube

Install the R package to read, subset, analyse and build your own cubes.

Resources: portal.geobon.org

github.com/EBVcube/EBVCube-format

EBVcube.github.io/ebvcube



Thank you

Questions, ideas, or critiques? Please let me know. 😊

Paul Miegel · Leipzig University / iDiv · June 2026

With thanks to the EBVCube team at iDiv for their guidance throughout the internship.

References

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